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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER B.TECH DEGREE EXAMINATION(2019 SCHEME), DECEMBER 2019

Course Code: MAT101

Course Name: LINEAR ALGEBRA AND CALCULUS

(2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

1

Determine the rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}$

(3)

2

If 2 is an eigen value of $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, without using its characteristic equation, find the other eigen values.

(3)

3

If $f(x,y) = xe^{-y} + 5y$ find the slope of $f(x_0)$ in the x-direction at (4,0).

(3)

4

Show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$, where $z = e^x \sin y + e^y \cos x$

(3)

5

Find the mass of the square lamina with vertices (0,0) (1,0) (1,1) and (0,1) and density function $x^2 y$

(3)

6

Evaluate $\int_0^{2\pi} \int_0^1 e^{-r^2} (x^2 + y^2) dr d\theta$ by changing to polar coordinates.

(3)

7

Test the convergence of the series $\sum_{k=1}^{\infty} \frac{k}{2k+1}$

(3)

8

Check the convergence of $\sum_{k=1}^{\infty} \frac{1}{k^4 \sqrt{2}}$

(3)

9

Find the Taylors series for $f(x) = \cos x$ about $x = \frac{\pi}{2}$ up to third degree term.

(3)

10

Find the Fourier half range sine series of $f(x) = e^x$ in $0 < x < 1$

(3)

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- PART B
Answer one full question from each module, each question carries 14 marks
- Module-I
- 11 a) Solve the system of equations by Gauss elimination method. (7)
- $x + 2y + 3z = 1$
 $2x + 3y + 2z = 2$
 $3x + 3y + 4z = 1$

- b) Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 4 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 3 \end{bmatrix}$. (7)
- 12 a) Find the values of λ and μ for which the system of equations
- $2x + 3y + 5z = 9$
 $7x + 3y - 2z = 8$
 $2x + 3y + \lambda z = \mu$
- has (i) no solution (ii) a unique solution and (iii) infinite solution

- b) Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$. Also write the diagonal matrix. (7)

- Module-II
- 13 a) Let f be a differentiable function of three variables and suppose that $w = f(x, y, z)$, show that $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$. (7)
- b) Locate all relative extrema of $f(x, y) = 4xy - y^4 - x^4$. (7)
- 14 a) Find the local linear approximation L to the function $f(x, y) = \sqrt{x^2 + y^2}$ at the point $P(3, 4)$. Compare the error in approximating f by L at the point $Q(3.04, 3.98)$ with the distance PQ . (7)
- b) The radius and height of a right circular cone are measured with errors of at most 1% and 4%, respectively. Use differentials to approximate the maximum percentage error in the calculated volume. (7)

- 15 a) Evaluate $\iint_R y dx dy$ where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$. (7)
- b) Use double integral to find the area of the region enclosed between the parabola $y = \frac{x^2}{2}$ and the line $y = 2x$. (7)

- 16 a) Evaluate $\int_0^1 \int_0^2 x^2 dz dy$ by reversing the order of integration (7)
- b) Use triple integrals to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$. (7)

Module-IV

- 17 a) Find the general term of the series $1 + \frac{1.2}{1.3}, \frac{1.2.3}{1.3.5}, \frac{1.2.3.4}{1.3.5.7}, \dots$ and use the ratio test to show that the series converges. (7)
- b) Test whether the following series is absolutely convergent or conditionally convergent $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt[4]{k(k+1)}}$ (7)

- 18 a) Test the convergence of $\frac{x}{1.2} + \frac{y^2}{2.3} + \frac{z^3}{3.4} + \dots + \frac{k^k}{k(k+1)}$ (7)
- b) Test the convergence of the series $\sum_{k=1}^{\infty} \frac{(k+1)!}{4!k!4^k}$ (7)

Module-V

- 19 a) Find the Fourier series of periodic function with period 2 which is given below $f(x) = \begin{cases} -x; & -1 \leq x \leq 0 \\ x; & 0 \leq x \leq 1 \end{cases}$ Hence prove that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$ (7)
- b) Find the half range cosine series for $f(x) = \begin{cases} kx & 0 \leq x \leq L/2 \\ k(L-x) & L/2 \leq x \leq L \end{cases}$ (7)

